Calibrated 6D vision for sequential tool manipulation

Task demonstration



00:00	00:03	00:08	00:16	00:19
Hammer return	**Moving screwdriver**	**Rack installation**	**Spanner regrasp**	**Hole-over-pin hanging**

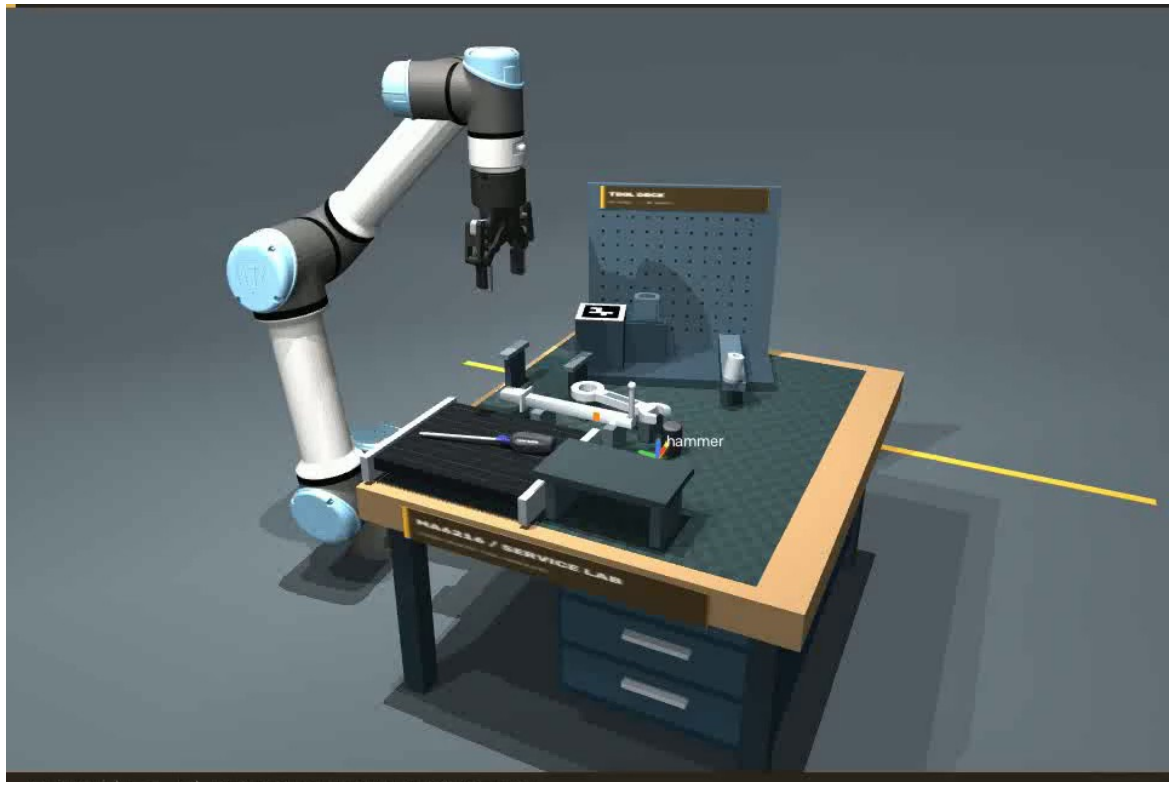
[Play standalone MP4 ↗]

Active task highlighted automatically · continuous 8× replay · 24.48 s

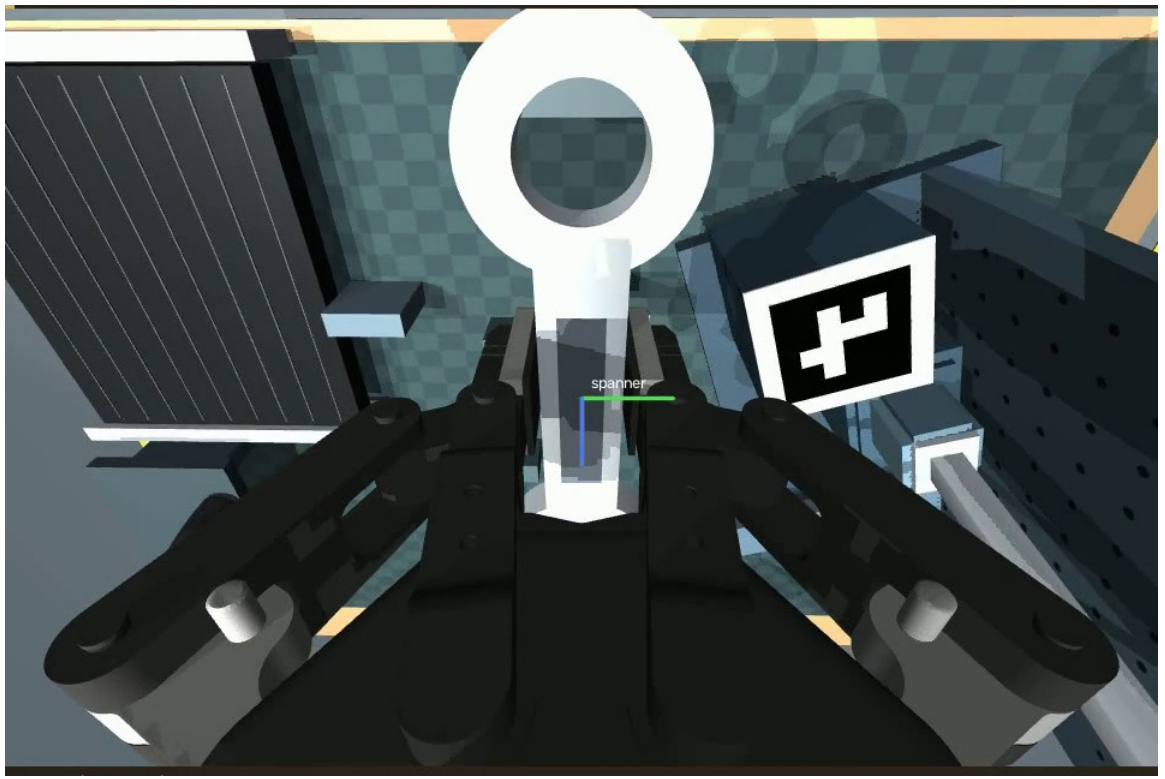
From visual observation to physical manipulation

Methods

RGB-D
Estimated intrinsics
→
Localization
Depth foreground
→
6D pose
FoundationPose
→
Robot target
Calibrated chain
→
Execution
IK + control



Workcell observation



Pose-guided grasp and reobservation

24 RGB-D reference views → reconstruction → registration / tracking. MegaPose and GraspNet were reviewed, not benchmarked.

Runtime observation: RGB, metric depth and predicted foreground

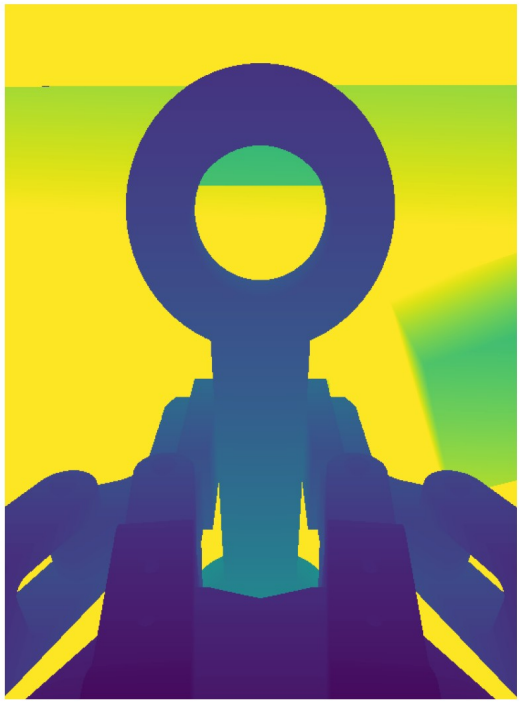
Traceable processing chain

a Calibrated RGB



Wrist RGB | t = 160.340 s

b Metric depth



Metric depth (display range clipped)

c Predicted foreground



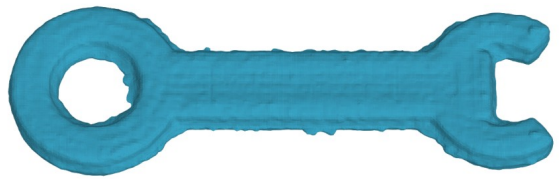
Predicted foreground mask

One wrist capture after regrasp: simulation t = 160.340 s, seed 400. Same crop across RGB, depth and predicted mask; no simulator IDs.

From reconstructed reference to estimated pose and execution

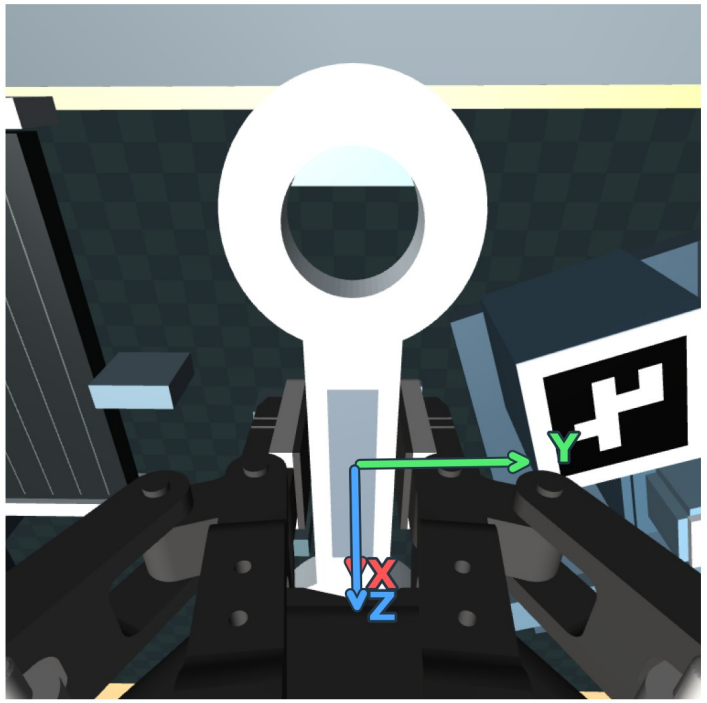
Traceable processing chain

a Reference mesh



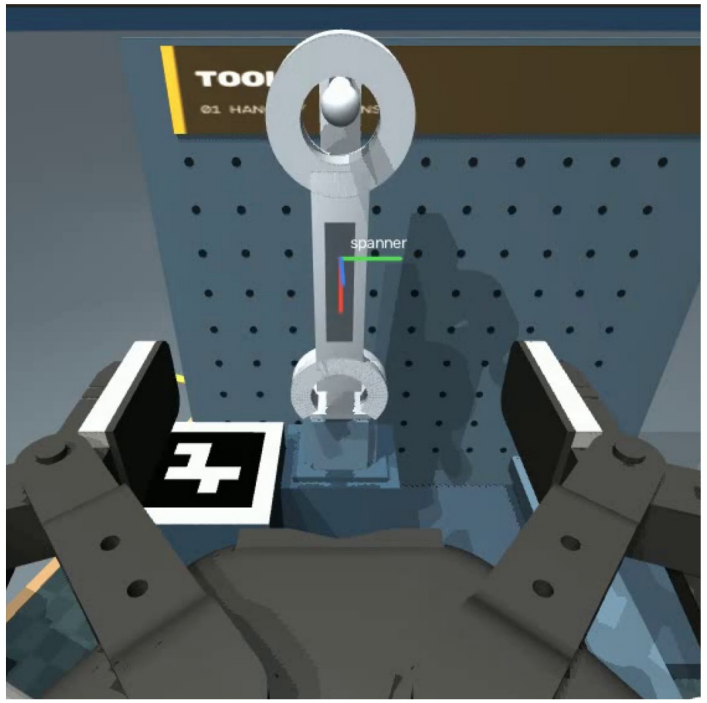
Prior reconstruction | 24 views

b Estimated pose



Same capture | estimated XYZ axes

c Released hanging



Later replay frame | t = 195.00 s

The reference is reconstructed beforehand from 24 views. Pose uses the preceding runtime observation; hanging is a later state of the same trial.

Image-based calibration and held-out validation

a Checkerboard observations

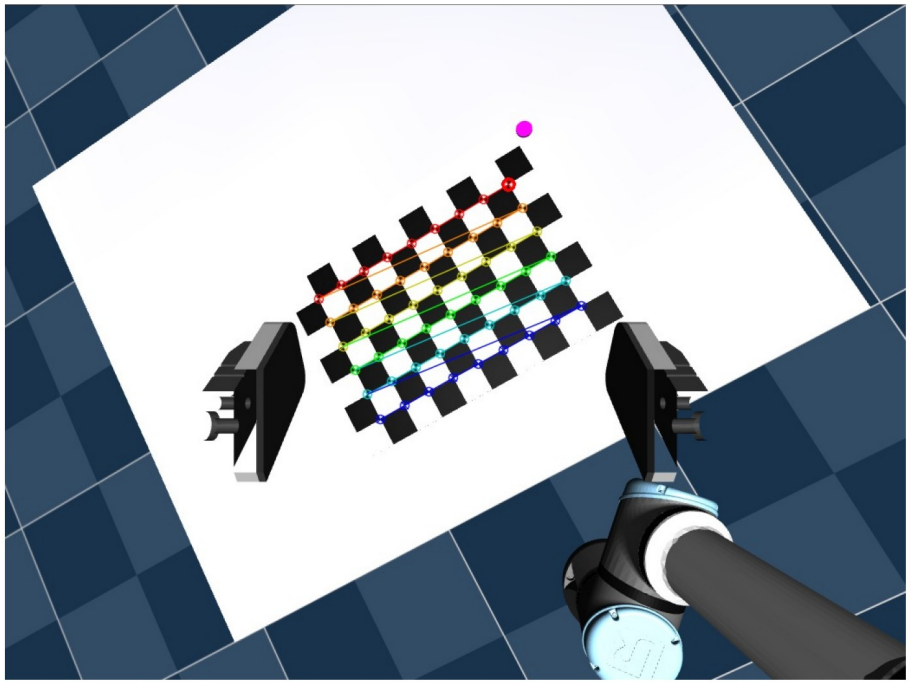
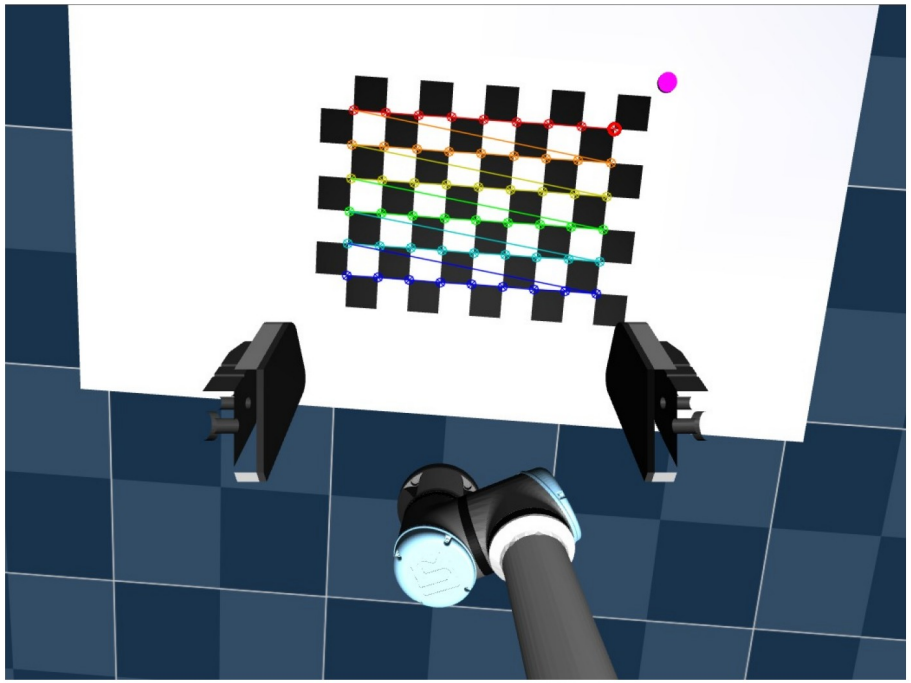


Image-based corner detection

b Held-out hand–eye view



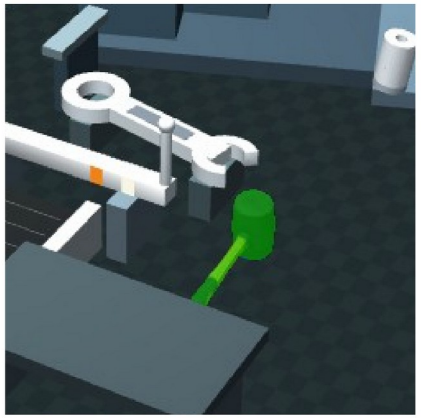
PnP + robot FK; frozen validation chain

Wrist intrinsics	Hand–eye transform	Frozen-chain reprojection
24 / 8 train–validation views	24 / 10 train–validation views	K and transformations fixed
0.02177 px RMSE	0.852 mm / 0.0227°	0.05497 px

Checkerboard detection + image PnP + robot FK. Simulator parameters are validation references; ideal pinhole images limit transfer.

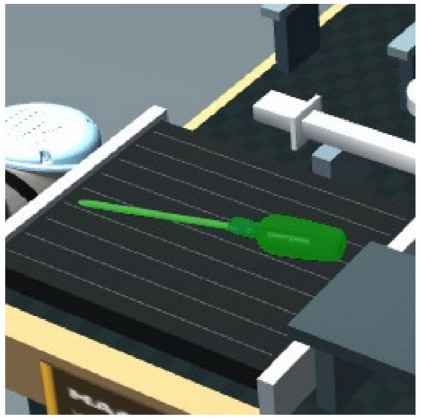
Localization: image masks, evaluation unit and assumptions

a Hammer



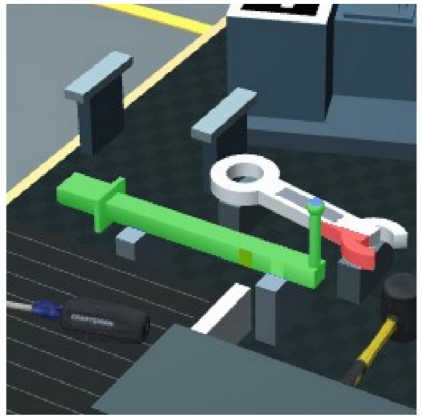
Nominal scene | seed 500

b Screwdriver



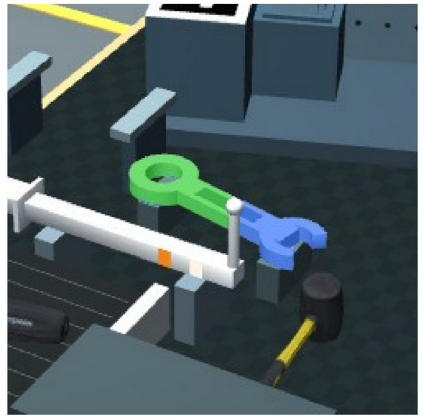
Nominal scene | seed 500

c Rack frame



Nominal scene | seed 500

d Spanner



Nominal scene | seed 500

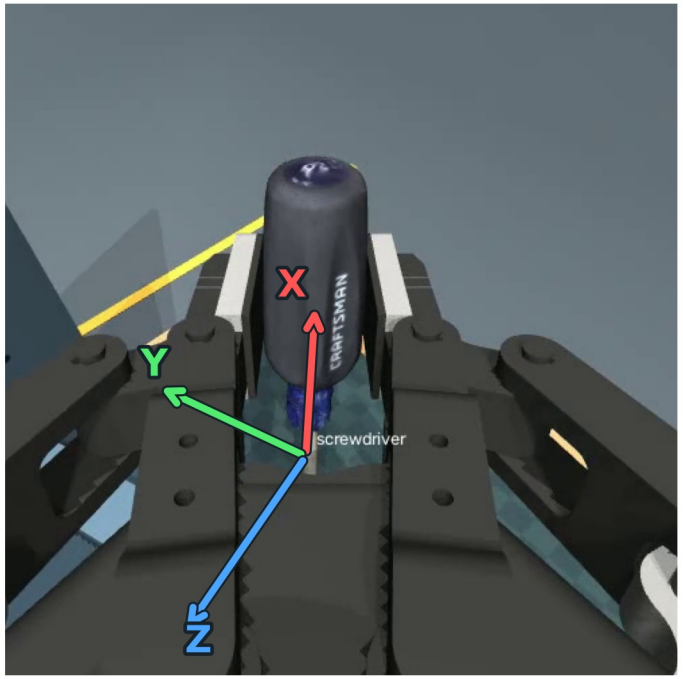
Detector	Independent evaluation	Boundary
Depth foreground subtraction Known pickup regions	50 scenes / 200 queries 10 scenes per tool–condition cell	Matched empty backgrounds Restricted occlusion / noise
Predictions saved first	160/160 positive queries: IoU ≥ 0.5 0/40 absent-target false positives	Wilson and scene-bootstrap intervals Not 200 robot executions

Nominal seed 500. Green: overlap; blue: missed visible target; red: extra foreground. Crops preserve original overlays.

Estimated object frames guide task-specific targets

a Screwdriver pose

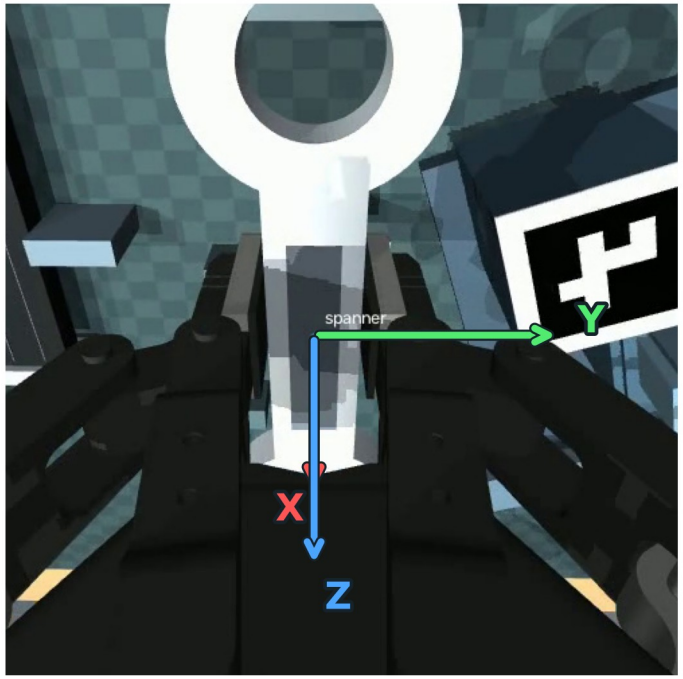
t = 54.40 s | wrist RGB



Estimated object frame

b Spanner pose after regrasp

t = 158.20 s | wrist RGB



Estimated object frame

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

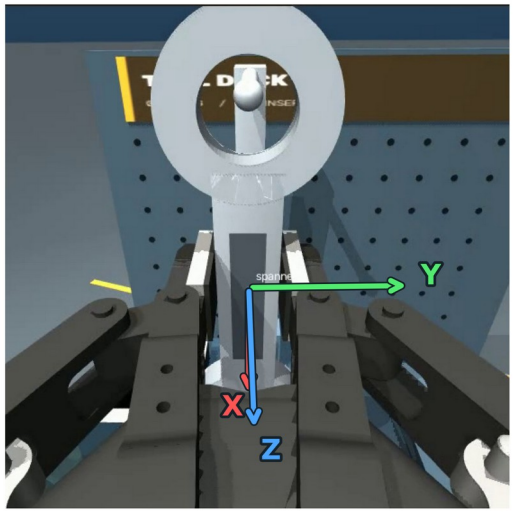
$$T_{BO} = T_{BG} \cdot T_{GC} \cdot T_{CO}$$

FK × hand–eye calibration × visual object pose

Reobserve after regrasp.

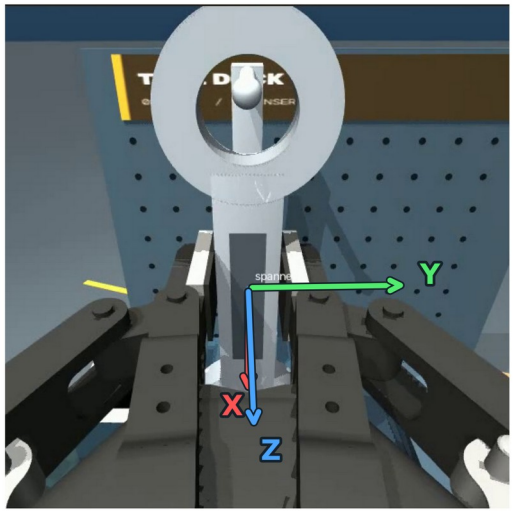
Visual feedback modes and bounded correction

a Before correction
t = 171.64 s | wrist RGB



Visual residual: 1.60°

b After reobservation
t = 172.76 s | wrist RGB



Visual residual: 0.74°

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

Observation schedule

Open loop
Initial acquisition only

Staged
Task-transition observations

Iterative
Repeated precontact correction

Common control limits

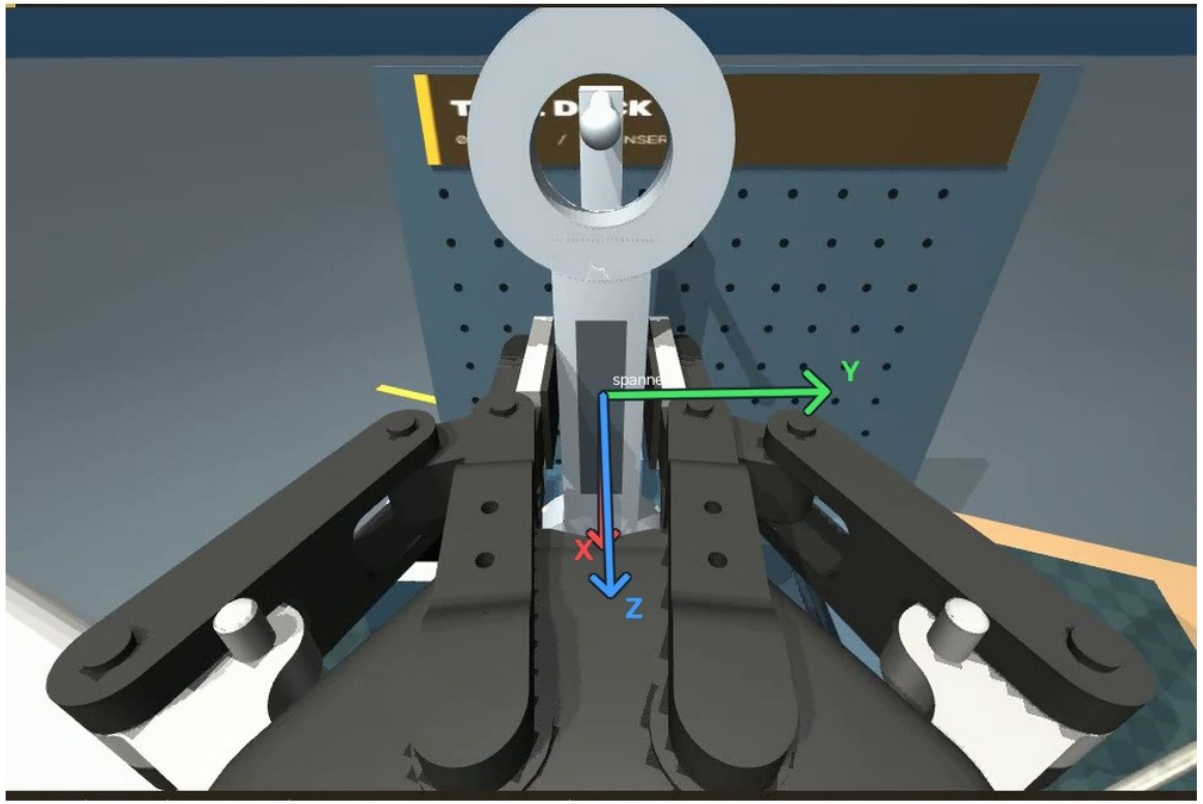
Step cap: 4 mm / 2°
Converged: two observations
within 1.5 mm / 1.5°

Estimated pose residual → bounded correction → reobserve. Same calibration, estimator and guards; discrete precontact control.

Logged visual feedback: one correction, three observations

Visual feedback: observe, correct, observe again

SEPARATE PERTURBED INTEGRATION RUN | 0.125x playback (8x slower) | recorded estimated pose



OBSERVATION LOG

CORRECTING

Observation 1 / 3

Position residual 1.13 mm

Angular residual 1.60°

Captured: 173.486 s

Available: 173.636 s

One corrective move commanded

0.758 mm / 1.037°

Convergence rule

Two consecutive observations within 1.5 mm and 1.5°

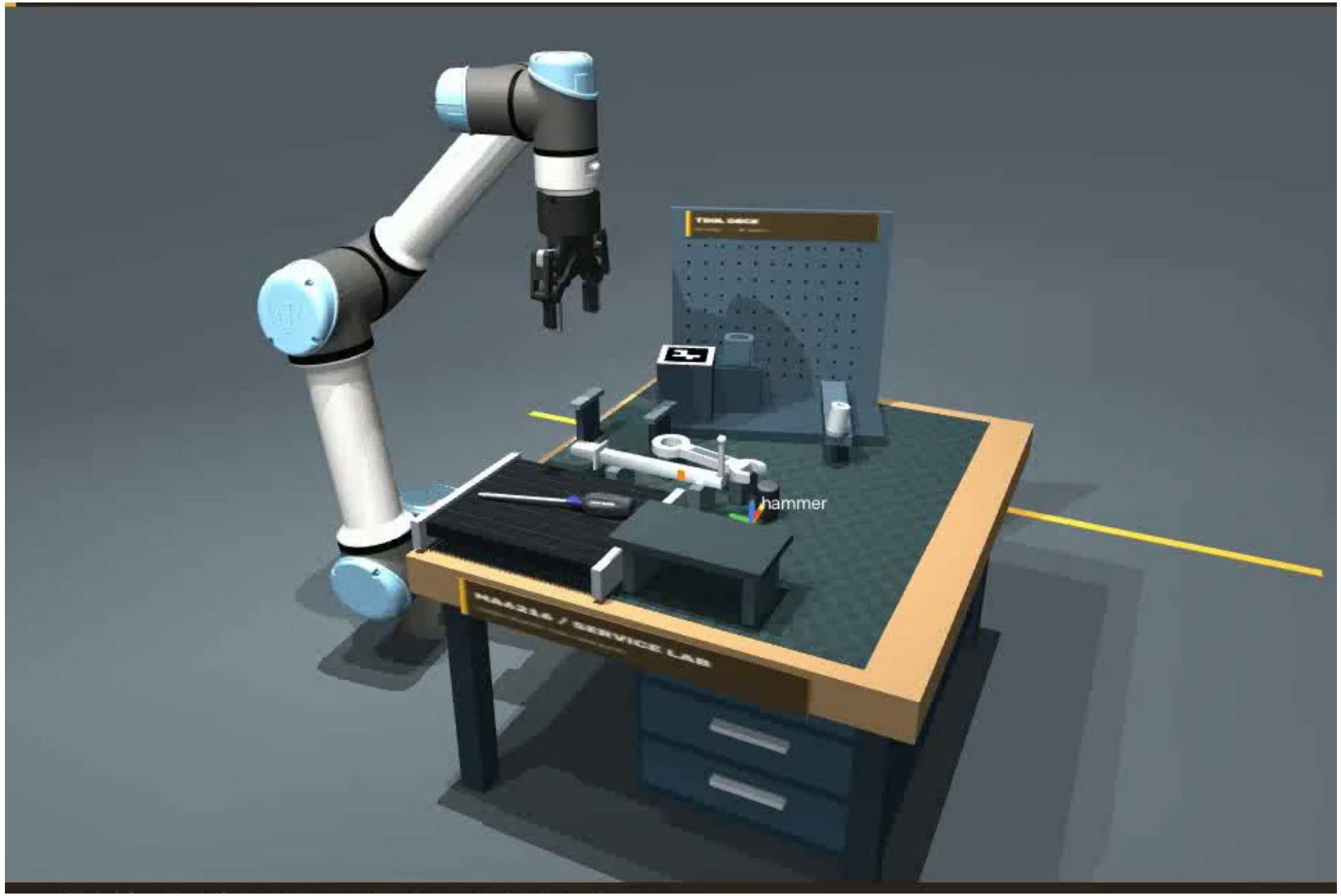
Source playback 172.125 s | Simulation time 174.127 s | Latest available measurement held between updates

Residuals compare estimated pose with the desired pose; not ground-truth error. This run is not pooled into the 48 trials.

Play standalone MP4 ↗

8× slower playback · logged values held between available observations

Paired evaluation in the complete workcell



Evaluation protocol

48 complete-task attempts

3 feedback modes
× 2 disturbance conditions
× 8 paired seeds

Outcome and uncertainty

All four physical checks must pass.

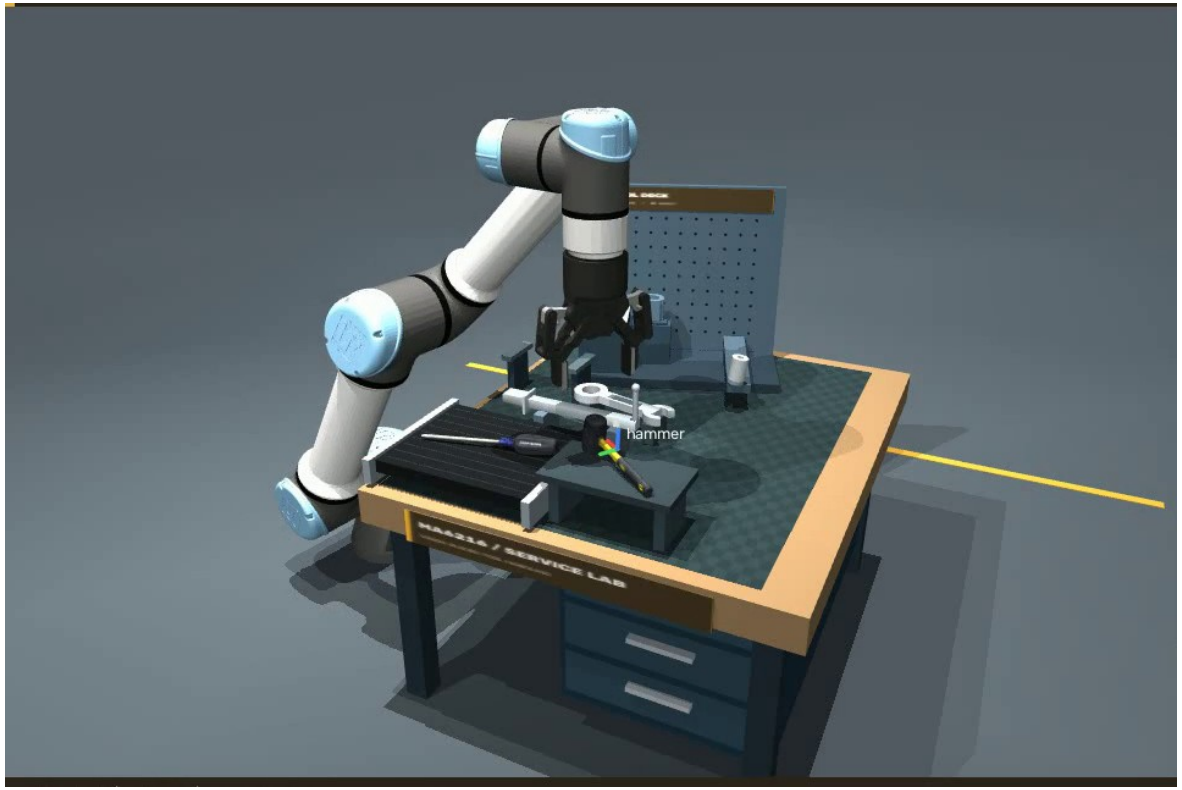
Wilson 95% intervals
Exact paired tests + Holm
Trial-level continuous summaries

Seeds 400–407 reused.
Paired replication, not a new holdout.
Early failures remain included.

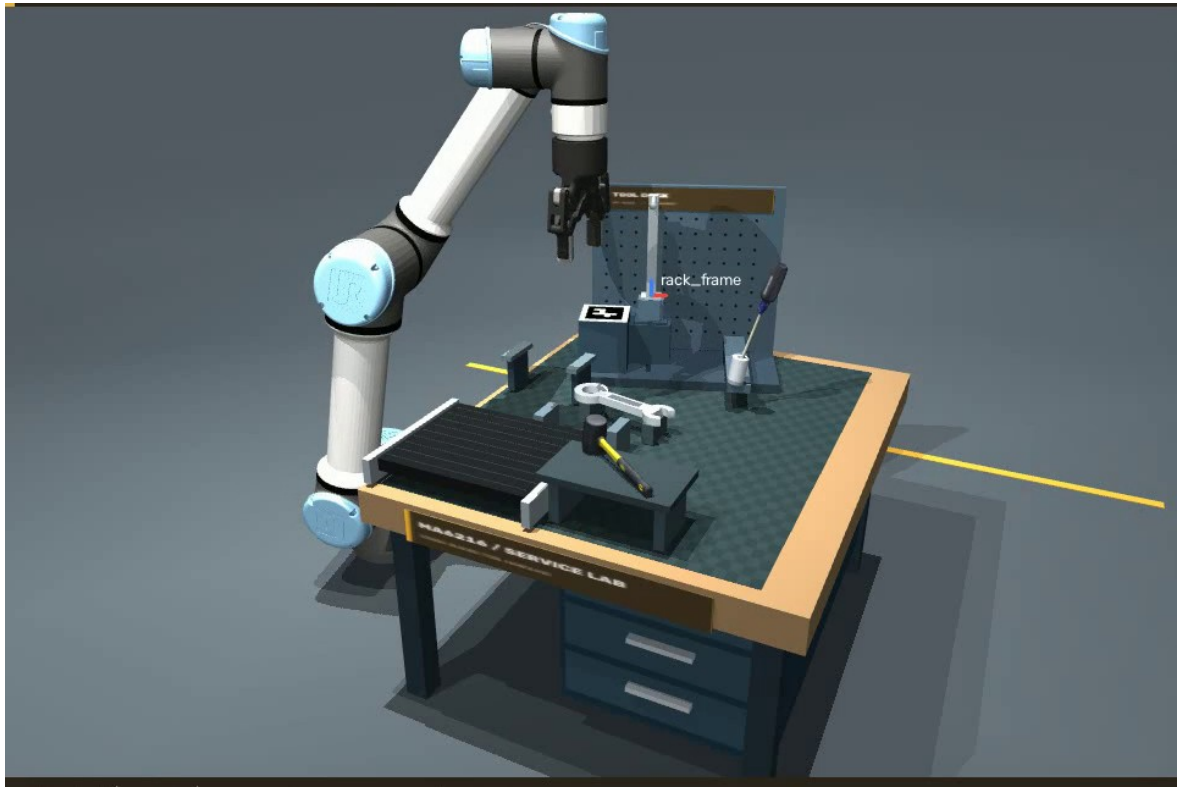
Physical return and independent rack installation

Representative successful run

a Released hammer placement · 25.00 s



b Released rack fixture · 134.40 s

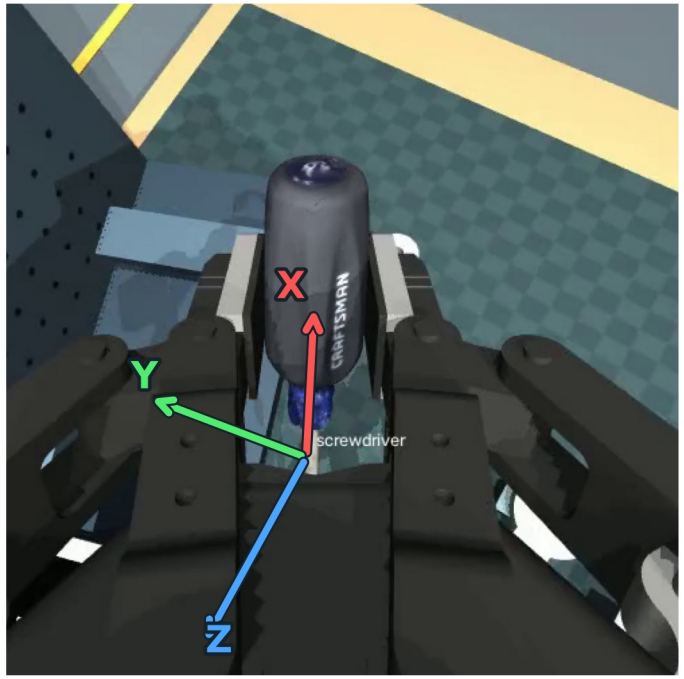


One continuous replay. The rack is installed as an independent object; released support and stability determine acceptance.

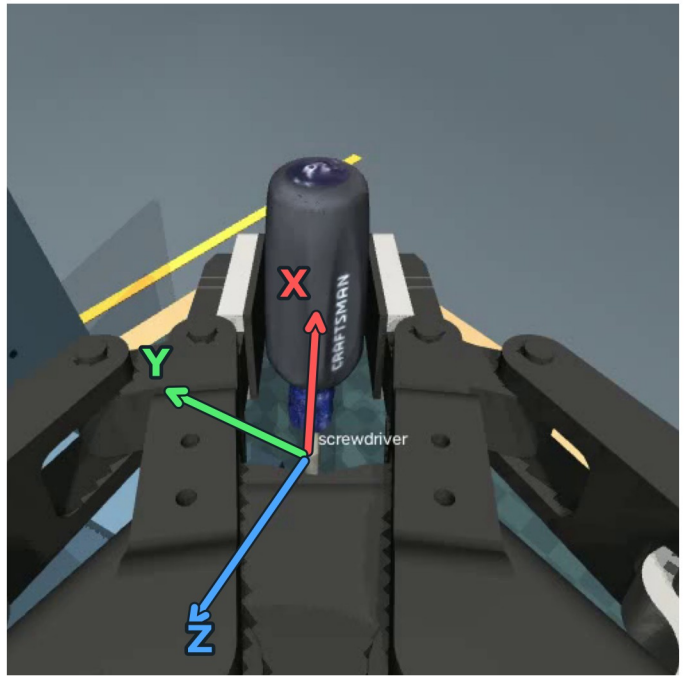
Tilted insertion: orientation, contact and released placement

Representative successful run

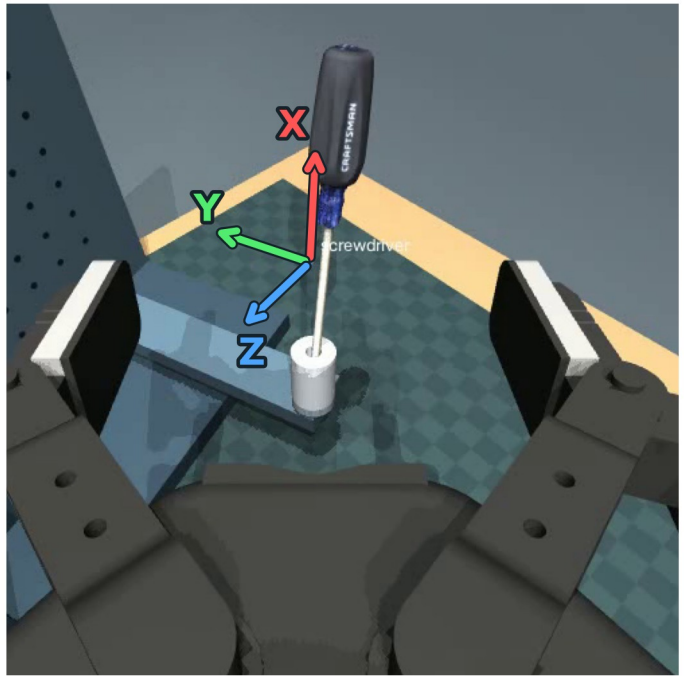
a Reorient
t = 50.00 s | wrist RGB



b Insert
t = 60.00 s | wrist RGB



c Release
t = 70.00 s | wrist RGB



Tool / gripper **Insertion region** **Placed tool**

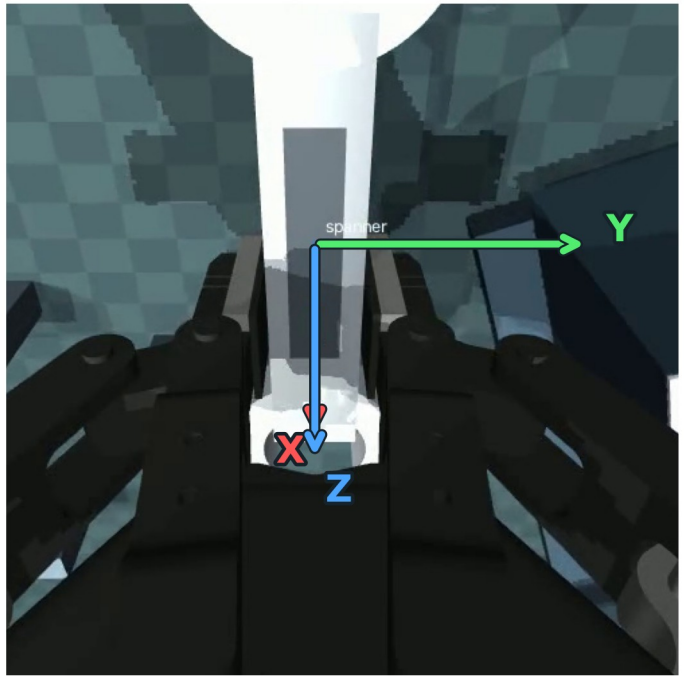
Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

Enlarged panels use matched pixel windows. Contact is partly occluded; physical acceptance comes from the recorded checks.

Physical regrasp: support, clearance and grasp reacquisition

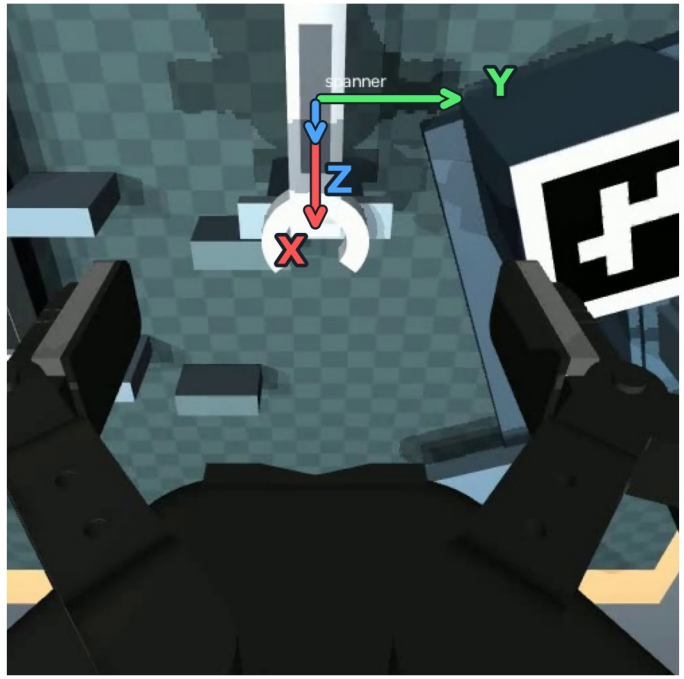
Representative successful run

a Place on support
t = 147.00 s | wrist RGB



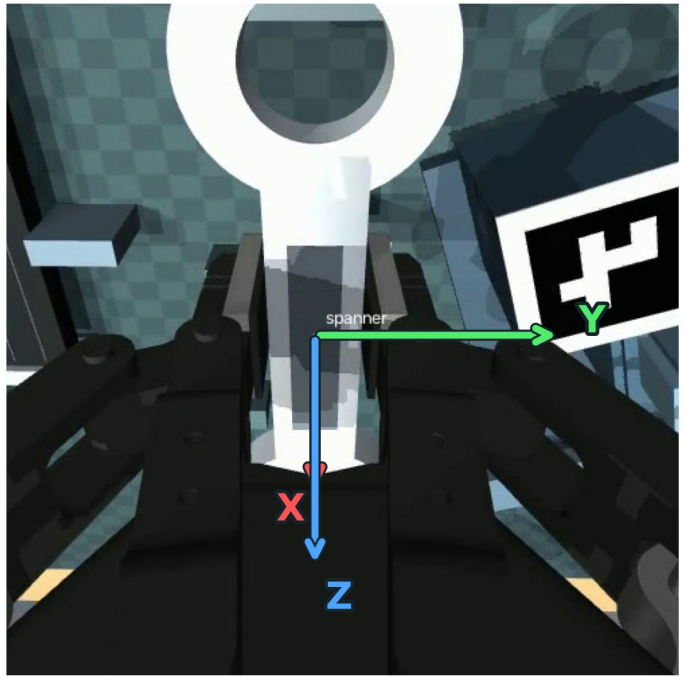
Original grasp

b Clear fingers
t = 150.40 s | wrist RGB



Open clearance

c Acquire new grasp
t = 158.20 s | wrist RGB



New grasp region

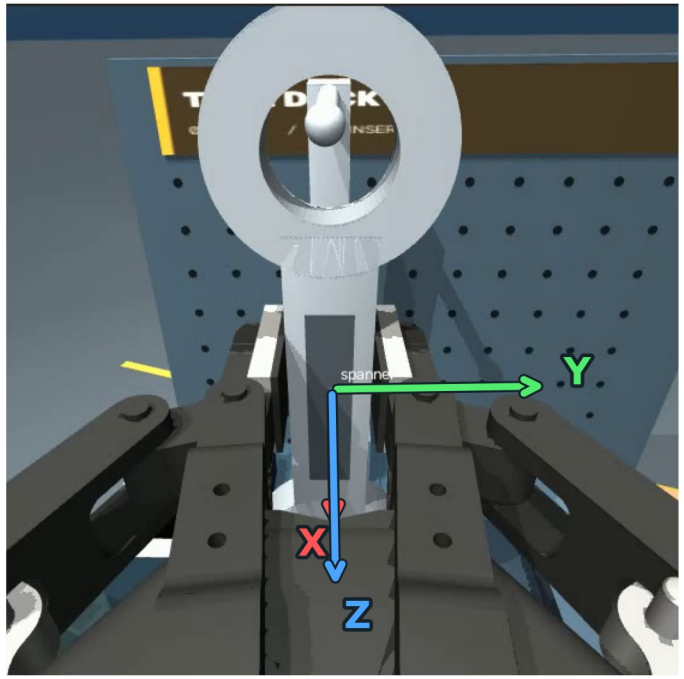
Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

One continuous trajectory. Enlarged views expose the grasp interface; new pose observation precedes reacquisition.

Hole-over-pin placement: local geometry and final support

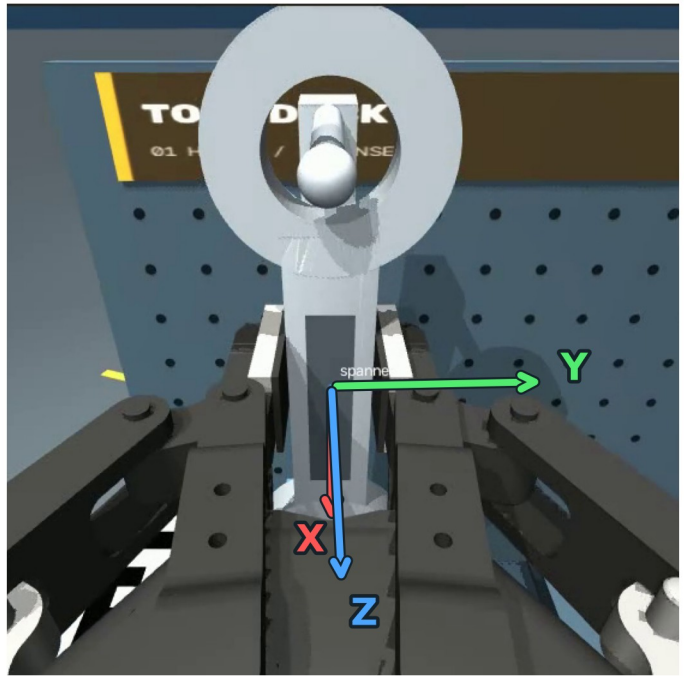
Representative successful run

a Align
t = 171.00 s | wrist RGB



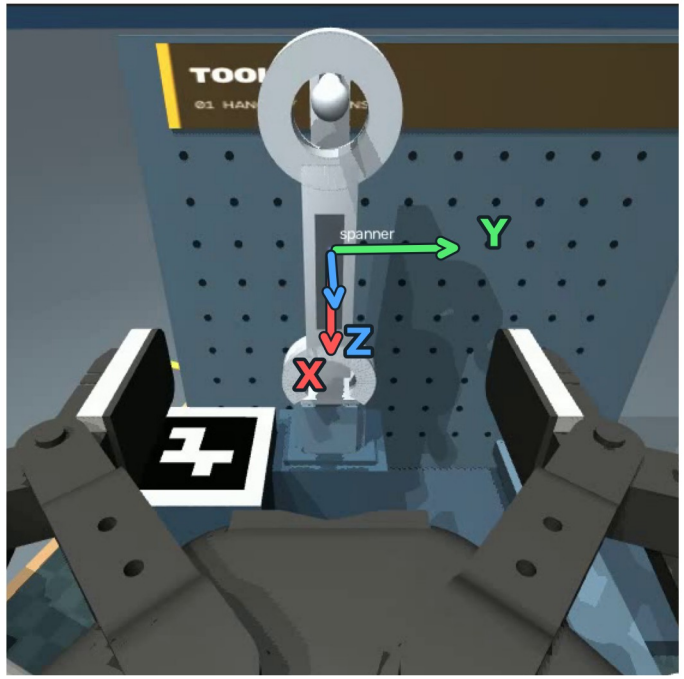
Ring / pin interface

b Thread
t = 179.00 s | wrist RGB



Pin inside ring

c Release
t = 195.00 s | wrist RGB



Released support

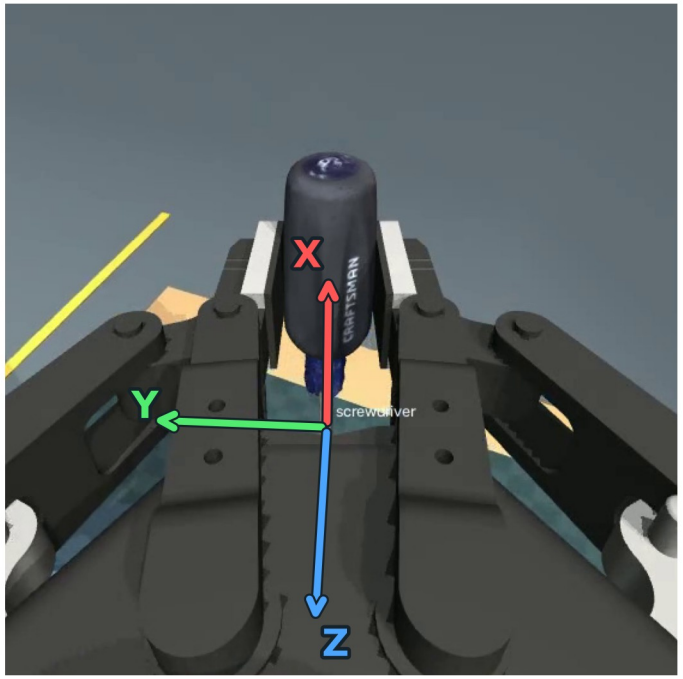
Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

Four terminal checks passed. Separate passive-stability audit: 30 s. Pose axes are visualization overlays, not measured contact forces.

Matched-seed comparison: insertion stop versus released placement

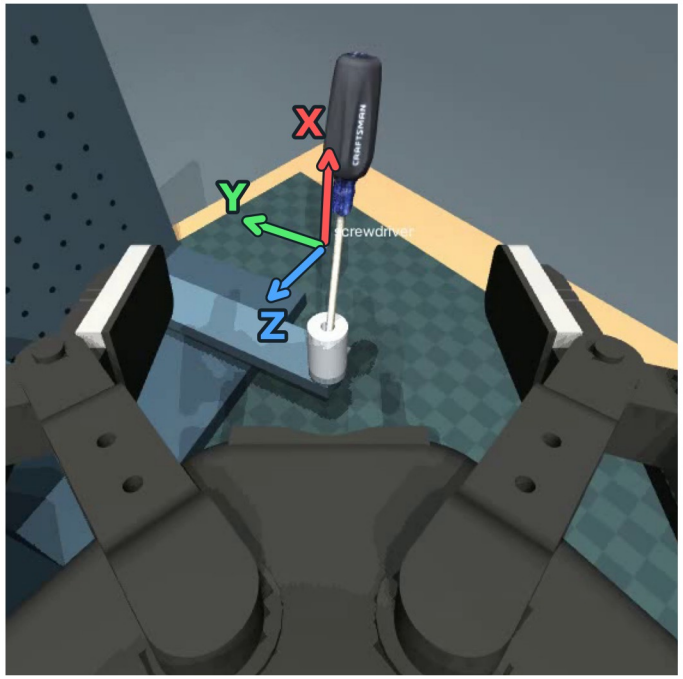
Controlled examples

a Open loop / seed 400
t = 57.96 s | wrist RGB



Insertion guard stop

b Iterative / seed 400
t = 70.00 s | wrist RGB



Released placement

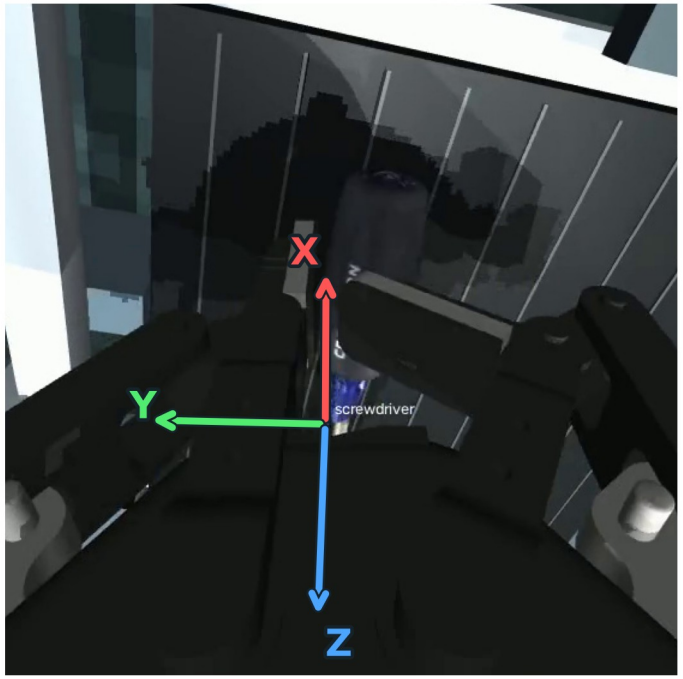
Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

Seed 400, no disturbance; endpoint times differ. A selected paired example does not establish statistical superiority.

Shared failure boundary: gripper closure before insertion

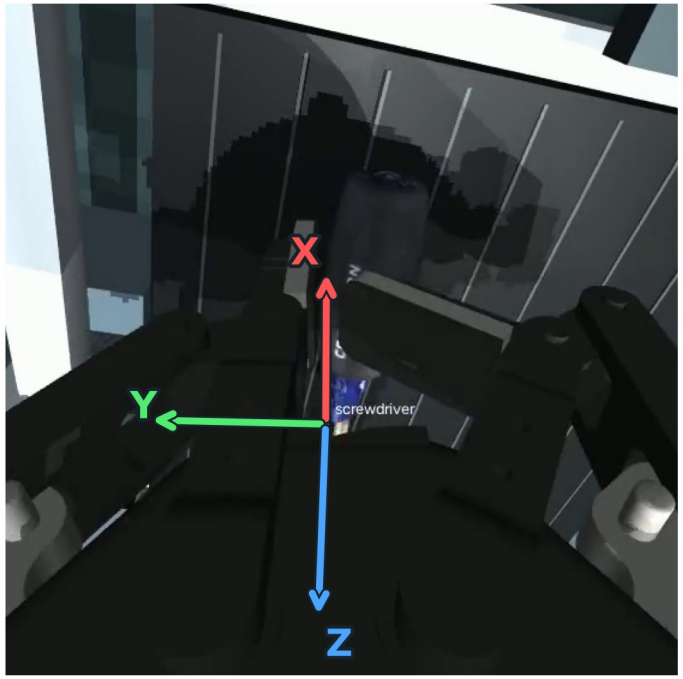
Controlled examples

a Staged / seed 401
t = 40.24 s | wrist RGB



Pickup guard stop

b Iterative / seed 401
t = 40.24 s | wrist RGB



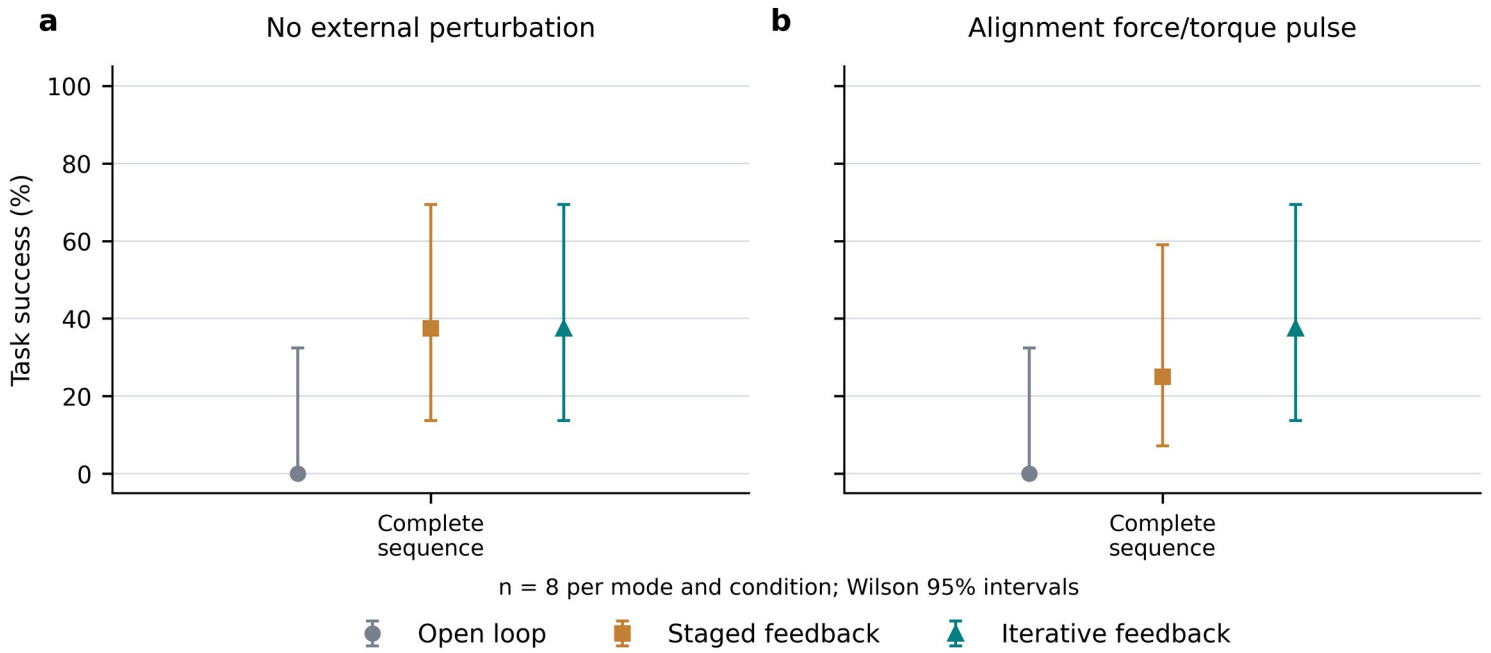
Same failure stage

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

Seed 401: both modes stop at gripper closure. Stage and guard attribution are from logs; all failed attempts remain in the analysis.

Whole-sequence results and statistical uncertainty

Quantitative results



Mode	None	Pulse
Open loop	0/8	0/8
Staged	3/8	2/8
Iterative	3/8	3/8

11/48 complete successes.
All four adjusted paired p-values: 1.0.
No statistically established superiority or equivalence in this sample.

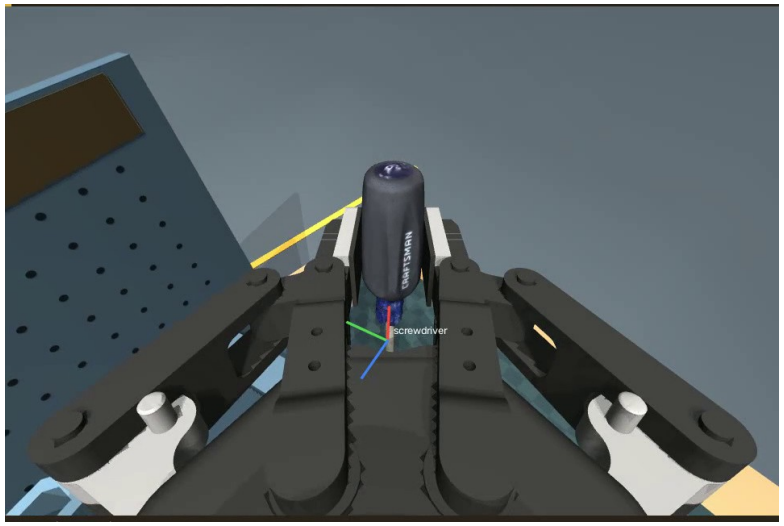
Interpretation

The selected successful trajectory demonstrates feasibility. Low complete-sequence reliability and wide intervals limit conclusions about feedback benefit. Early failures remain part of the system-level outcome.

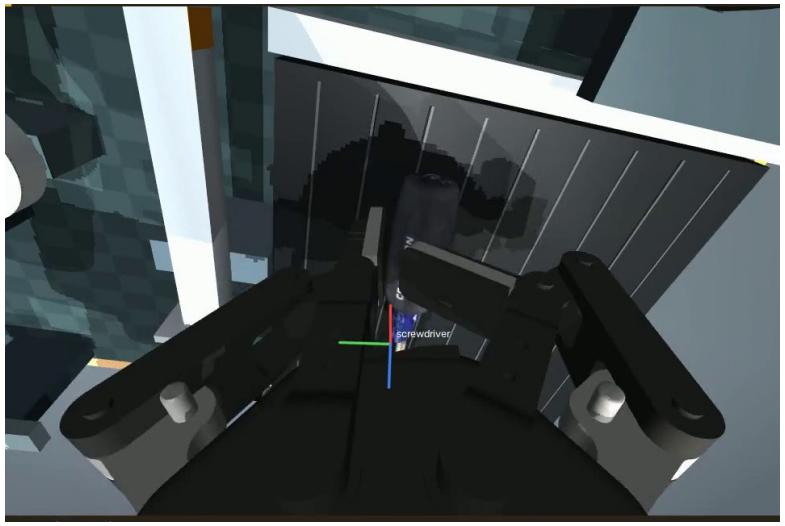
Pose accuracy and the physical failure boundary

Error decomposition

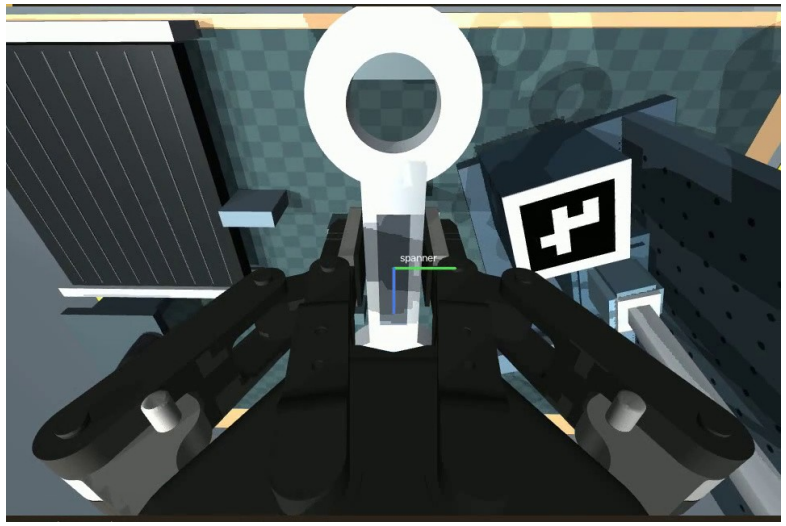
Visual pose / initial acquisition



Pickup contact guard



Grasp reliability after reobservation



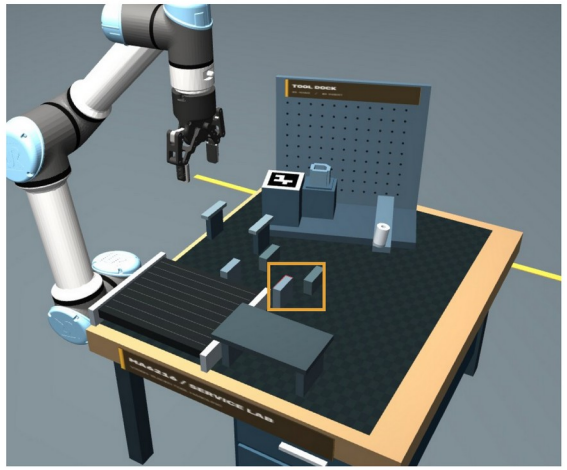
Initial pose metric	Mean
Translation / rotation	2.60 mm / 1.44°
ADD / ADD-S	3.17 / 2.29 mm

Failure category	Attempts
Pickup contact / hammer planning	24 / 6
Insertion guard / spanner grasp loss	6 / 1

Initial pose error, TCP tracking and final placement are different endpoints. 500 Hz physics is not visual feedback frequency.

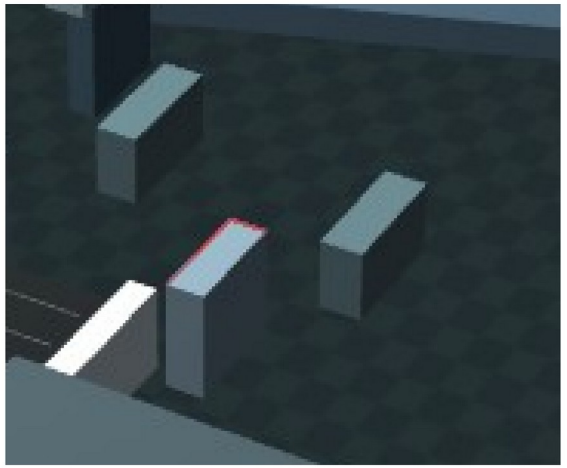
Generalization boundary: background-dependent localization

a Target-absent scene



Separate background-mismatch diagnosis

b Magnified false-positive region



Red pixels: predicted foreground without target

Observed limitation

Absent rack target, but mismatched background geometry produces a false-positive region.

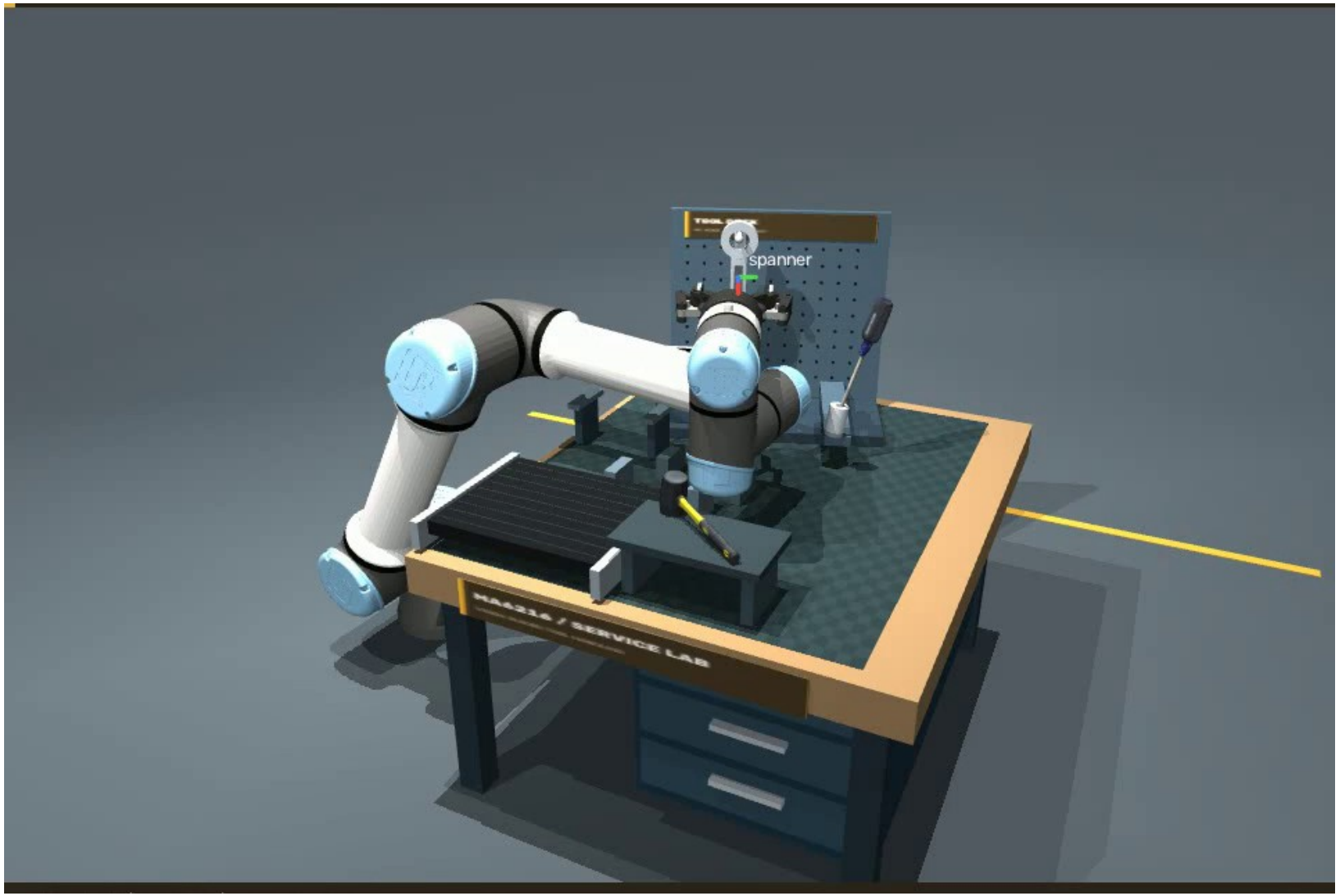
The zoom is from the same original image.

Assumption	Impact on interpretation
Known pickup stations and paired backgrounds	Does not establish arbitrary-clutter recognition
Synthetic scans, ideal FK / depth, rigid mounts	Does not establish real-camera or hardware reliability
Small reused-seed sample	Broader unseen-scene trials remain necessary

Separate development diagnosis; not pooled into the 50-scene localization audit or the 48 robot attempts.

What the integrated system demonstrates

Discussion



Demonstrated

Independent calibration
Visual 6D pose integration
Physical regrasp and assembly
Auditable feedback comparison

Next priorities

Improve pickup contacts.
Test changed backgrounds.
Use unseen seeds and external baselines.
Validate on hardware.

High reliability and statistical superiority are not established.

References and reproducibility

Appendix

Selected methods

Wen et al. FoundationPose. CVPR, 2024.

Zhang. A flexible new technique for camera calibration. TPAMI, 2000.

OpenCV. Camera and hand–eye calibration documentation.

Chaumette & Hutchinson. Visual servo control, Part I. RAM, 2006.

Todorov et al. MuJoCo. IROS, 2012.

Reviewed alternatives

Labbé et al. MegaPose. arXiv:2212.06870, 2022.

Fang et al. GraspNet-1Billion. CVPR, 2020.

Repository evidence

Calibration images + robot FK

Predicted and evaluation masks

48 complete trial summaries

Original successful and failed replays

15 annotated image plates (PDF / SVG)

Frame indices, crop boxes and source hashes

Full references: docs/LITERATURE_AND_SELECTION.md

Experiment report: visual-feedback-calibrated-v1

Simulation images illustrate observed behavior; quantitative conclusions use all predefined attempted trials.